SDUHSD CAREER TECHNICAL EDUCATION

Advanced Engineering Technology

Campus: Canyon Crest Academy
CTE Industry Sector: Engineering and Architecture
CTE Pathway: Engineering Technology
Course Sequence: Capstone
Prerequisites: Introduction to Engineering & Digital Electronic
OR
Exploring Computer Science
OR
Human-Computer Interaction (HCI) Design
OR
APCSA

10 credits, UC Approved Area G
Recommended Pathway Sequence:

Course 1 - Introductory	Course 2 - Concentrator	Course 3- Capstone	Enrichment
Introduction to Engineering Techniques	Digital Electronics	Advanced Engineering Technology	Robotics & Engineering Clubs

Brief Course Description:

- Standards for Engineering and Architecture
- http://www.corestandards.org/ELA-Literacy/RST/11-12/
- http://www.nextgenscience.org

Course Syllabi:
CCA

Grading: A variety of assessments will be used including journal entries, projects and self and peer assessments. For example students will be assessed on their journal entries which include project plan and schedule updates. Another example assessment includes demonstration of the mechanical, electrical, and control mechanisms related to a proposed project. Another example assessment includes a student self-evaluation of the design of his or her proposed project.

Workload: (High, Medium, Low)

Class setting: This relates directly to how much the student is expected to work independently or in a group requiring self-initiated work, problem solving, etc versus and highly guided and structured class).
(Highly independent, Moderately Independent, Structured). Hands-on projects accompany each unit.
*Each year all CTE courses are reviewed by industry professionals within the pathway to ensure the most current, relevant experience for our students

Textbooks or Supplemental Materials for this course: Supplemental Materials provided electronically.

Software, program knowledge specific to this course: Autodesk Inventor, Autodesk Revit, LoggerPro, MDsolids, Scilab, Eclipse, Android Studio, Eagle Board Designer

Industry Connections: Every February - students showcase projects with local businesses and community members

Units of Study, Large Projects, Work-based Learning Connections.

1. Introduction to the Design and Development Process
2. Problem Identification (Proposal - Problem to solve)
3. Research (How others have solved problem, patent searches)
4. Decision Process and Requirements (Proposed project, Systems requirements document, Derived requirements document, Acceptance test plans)
5. Agile Processes (Project plan)
6. Design (3D model, schematics, simulation)
7. Build (Fabrication of proposed project)
8. Test (Verification of project against acceptance test plan)
9. Presentation (Demonstration, Report)

The following Model Curriculum Standards connected to this course:

Standards for Career Ready Practice

1. Apply appropriate technical skills and academic knowledge. 4. Apply technology to enhance productivity. 5. Utilize critical thinking to make sense of problems and persevere in solving them. 10. Demonstrate creativity and innovation.

Framework Concept	Engineering Technology
Introduction to the Design and Development Process	B1.0 Communicate and interpret information clearly in industry-standard visual and written formats. B9.0 Understand the fundamentals of systems and market influences on products as they are developed and released to production.
Problem Identification	B2.0 Demonstrate the sketching process used in concept development. B4.0 Understand the concepts of physics that are fundamental to engineering technology. B6.0 Employ the design process to solve analysis and design problems. B7.0 Understand industrial engineering processes, including the use of tools and equipment, methods of measurement, and quality assurance. B10.0 Design and construct a culminating project effectively using engineering technology.
Research	B9.0 Understand the fundamentals of systems and market influences on products as they are developed and released to production. B10.0 Design and construct a culminating project effectively using engineering technology.
Decision Process and Requirements	B4.0 Understand the concepts of physics that are fundamental to engineering technology. B5.0 Understand how the principles of force, work, rate, power, energy, and resistance relate to mechanical, electrical, fluid, and thermal engineering systems. B7.0 Understand industrial engineering processes, including the use of tools and equipment, methods of measurement, and quality assurance. B10.0 Design and construct a culminating project effectively using engineering technology.
Agile Processes	B1.0 Communicate and interpret information clearly in industry-standard visual and written formats. B6.0 Employ the design process to solve analysis and design problems. B7.0 Understand industrial engineering processes, including the use of tools and equipment, methods of measurement, and quality assurance.
Design	B2.0 Demonstrate the sketching process used in concept development. B3.0 Identify the fundamentals of the theory, measurement, control, and applications of electrical energy, including alternating and direct currents. B4.0 Understand the concepts of physics that are fundamental to engineering technology. B5.0 Understand how the principles of force, work, rate, power, energy, and resistance relate to mechanical, electrical, fluid, and thermal engineering systems. B6.0 Employ the design process to solve analysis and design problems.
Build	B4.0 Understand the concepts of physics that are fundamental to engineering technology.

	B5.0 Understand how the principles of force, work, rate, power, energy, and resistance relate to mechanical, electrical, fluid, and thermal engineering systems.
Test	B3.0 Identify the fundamentals of the theory, measurement, control, and applications of electrical energy, including alternating and direct currents.
Presentation	B1.0 Communicate and interpret information clearly in industry-standard visual and written formats. B11.0 Understand the methods of creating both written and digital portfolios.